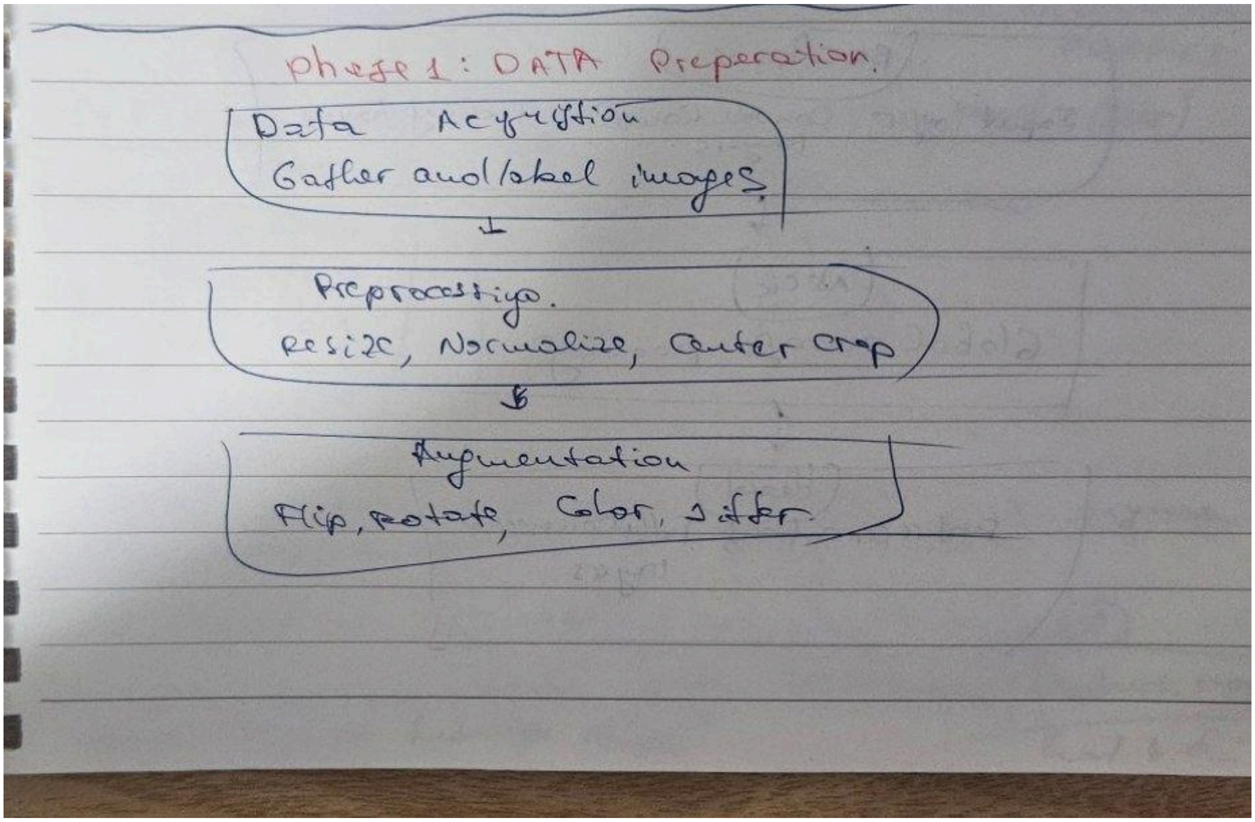


Phase 1: DATA Preperation.

Data Acquisition
Gather and label images.

Preprocessing.
Resize, Normalize, Center crop

Augmentation
Flip, rotate, Color, Jitter

Tayyor datasetlar mavjud:

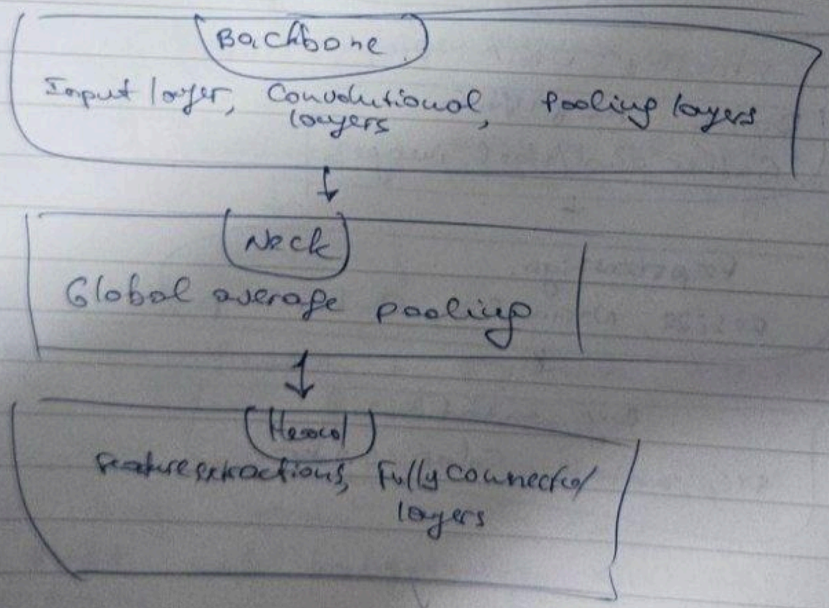
CIFAR 10 -> 10 ta classdan iborat. 32x32 pixels.
Car, bird, cat, deer, dog frog, horse, ship, truck kabilarni classify qilish uchun ishlata olamiz.

MNIST -> 10 ta class 0 dan 9 gacha 28x28 rasmlar.
qo'lda yozilgan sonlar. grayscale

Torchvision datasets -> PyTorchdagi kutubxona bo'lib, mashhur datasetlarni yuklash va ishlatishni osonlashtiradi yuqoridagilarni.

Hugging Face datasets -> turli sohalardagi datasetlarni topish, yuklash va qayta ishlash uchun kutubxona hamda platforma.

Phase 2: Model Architecture


Backbone
Input layer, Convolutional layers, Pooling layers
Neck
Global average pooling
Head
Feature extractions, Fully connected layers

o'simlik bargi bo'yicha

Input layer:

Height H = 224
Weight W = 224
Channels C = 3 R, G, B.

$224 \times 224 \times 3 = \boxed{150528}$ pixels

Normalization → $x_{new} = \dfrac{x_{original}}{255}$ [0,1]

pytorchda (3, 224, 224). batch size → 32 bo'lsa,
yakuniy natija = [32, 3, 224, 224].
bir vaqtda modelga beriladigan rasmlar soni.

Input → [32, 3, 224, 224]

Backbone

Convolutional layer: Rasm bo'ylab filtr (kernel) harakatlanib ma'lum belgilarni aniqlash jarayoni.

CNN: dastlabki qatlamlari: qirralar, chiziqlar, rang o'zgarishi.
o'rta qatlam: dog'lar, barg tomirlari;
chuqurroq qatlam: kasallikka xos murakkab pattern.

5×5 grayscale image bor. 3×3 kernel (filtr).

birinchi 3×3

mos qiymatlar ko'paytiriladi

Input

$$X = \begin{bmatrix} 1 & 2 & 0 & 1 & 3 \\ 4 & 1 & 2 & 2 & 0 \\ 3 & 2 & 1 & 0 & 1 \\ 0 & 1 & 3 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 \end{bmatrix}$$

kernel

$$K = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & -1 \\ 1 & 0 & -1 \end{bmatrix}$$

Harbir joydagi qiymatni kernelning mos qiymatiga ko'paytiramiz

1-hudud → $(1+0+0)+(4-0-2)+(3-0-1)=\boxed{5}$

~~Jarayon davom etadi~~ va natijada feature map hosil bo'ladi
kernel keyingi hududga siljiydi

Feature map o'lchamini hisoblash.

$$\boxed{\frac{N+2P-k}{S}+1}$$

N - input o'lcham → 5×5.
P - padding → atrofga qo'shiladigan pixels → 0
k - kernel o'lcham → 3×3.
S - stride → 1. kernelning siljish qadami

feature mapga Activation Function qo'llaniladi.

ReLU(x) = max(0, x) nonlinearity qo'shadi.

$$\frac{5+2\cdot0-3}{1}+1=3$$

feature map:

1) $(1+0+0)+(4+0-2)+(3+0-1)=1+2+2=5.$
2) $(2+0-1)+(1+0-2)+(2+0+0)=1-1+2=2.$
3) $(0+0-3)+(2+0+0)+(1+0-1)=-3+2+0=-1$
4) $(4+0-2)+(3+0-1)+(0+0-3)=2+2-3=1$
5) $(1+0-2)+(2+0+0)+(1+0-2)=-1+2-1=0.$
6) $(2+0+0)+(1+0-1)+(3+0-2)=2+0+1=3$
7) $(3+0-1)+(0+0-3)+(2+0-1)=2-3+1=0$
8) $(2+0+0)+(1+0-2)+(0+0-1)=2-1+1=0$
9) $(1+0-1)+(3+0-2)+(1+0+0)=0+1+1=2.$

$$\begin{bmatrix}5 & 2 & -1\\1 & 0 & 3\\0 & 0 & 2\end{bmatrix} \Rightarrow \text{ReLU} \Rightarrow \begin{bmatrix}5 & 2 & 0\\1 & 0 & 3\\0 & 0 & 2\end{bmatrix}$$

Pooling layer: Pooling feature mapning o'lchamini kichraytiradi. Muhim signallarni saqlab qolishga yordam beradi.

maxpooling → maximalini oladi.
average pooling → o'rtachasini.

2×2 pooling, stride = 1 → qadam.

max pooling

1) max (5, 2, 1, 0) = 5.
2) max (2, 0, 0, 3) = 3.
3) max (1, 0, 0, 0) = 1
4) max (0, 3, 0, 2) = 3

$$\left.\begin{matrix}\\ \\ \\ \\ \end{matrix}\right\} \begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix} \quad \underline{2\times2}$$

average =>

1) 8 / 4 = 2
2) 5 / 4 = 1.25
3) 1 / 4 = 0.25.
4) 5 / 4 = 1.25

$$\left.\begin{matrix}\\ \\ \\ \\ \end{matrix}\right\} \begin{bmatrix} 2 & 1.25 \\ 0.25 & 1.25 \end{bmatrix} \quad 2\times2$$

Pooling uchun => $\frac{3 + 2 \cdot 0 - 2}{1} + 1 = \boxed{2}$

xuddi shu jarayon filterlar soniga qarab takrorlanadi.
Agar 16 ta filter ishlatsak natija
(2 × 2 × 16) bo'ladi.

Neck (Global average pooling).
=> Head va Backbone ni o'rtasida turadi.
=> Yakuniy qarordan oldin ma'lumotni tayyorlaydi. (tartiblaydi.

Global average pooling
Convert feature map → single vector.
feature map (H × W × C) → vector (1 × 1 × C)

① → Convolutional → ReLU → GAP → Flatten → Fully Connected

$(5+0+2) + (1+0+3) + (0+0+2) / 9 = 7+4+2 / 9 = 13/9 = \boxed{1.44}$

3×~~3 × 16 → 1×1×16. — 16 → filterlar [illegible]~~

3×3×1 → 1×1×1 → $\boxed{1.44}$.

2. Convolutional → ReLU → MaxPooling → GAP → Flatten → FC

$\begin{bmatrix} 5 & 3 \\ 1 & 3 \end{bmatrix} \rightarrow 5+3+1+3/4 = 12/4 =$ ③

$2 \times 2 \times 1 \rightarrow 1 \times 1 \times 1 \rightarrow$ ③.

16 ta filter tan so'ng $1 \times 1 \times 16$ –
16 ta sondan iborat vector

Head

Fully Connected Layer.
Dense layer → munosabatlarni o'rganadi va qaror chiqaradi.

Softmax Activation:
outputlarni ehtimollikga almashtiradi.
/

Backbone → umurtqa pog'onasi → asosiy tayanch
feature extractor.
Neck → Bo'yin → Bog'lovchi, ma'lumotni uzatadi
Head → Bosh → Qaror qabul qiladi.